XRPL Validator — Daily Engineering Report · 2026-04-15

Period: 2026-04-15 00:00 to 24:00 US Eastern (2026-04-15 04:00 to 2026-04-16 04:00 UTC). Generated 2026-10-21 00:47 EDT by report generator v3.0. Sources: the XRP Ledger (ledger headers and amendments, read from a reference rippled node), the validator's signing log and receipt logs, the deployment pipeline's gate, soak and health records, the git history of xrpl-validator and the XRPL Rust SDK, and data.xrpl.org. Scope: XRPL work only.

Revised on 2026-10-21 in report format v3; the original is kept in this repository's history. Built only from records that exist for this day. Available: data.xrpl.org agreement scores. Not available: ledger headers (the reference node holds ledgers from 2026-10-04 22:52 UTC); signing records (they begin 2026-09-19); Step 1 receipt records (they begin 2026-08-29); Steps 2–5 receipt records (they begin 2026-10-09); deployment records (they begin 2026-09-11); xrpl-validator merge records (they begin 2026-07-04); amendment data.

Status

Not assessed — no signing records for this day.

Criteria. Normal: coverage outside planned maintenance at or above the 99.9% objective, every validation agreed with the network's validated ledger, and no engine receipt is open. Degraded: coverage below the objective, or an open engine receipt. Impaired: any validation that disagreed with the network, or coverage below 90%.

Executive summary

- **Validator:** no signing records for this day (they begin 2026-09-19).
- **Native engine:** no receipt records for this day (they begin 2026-08-29).
- **Changes:** 0 deployments, 32 commits (0 with a recorded merge into main; xrpl-validator merge records begin 2026-07-04).

Service levels

No signing records for this day (they begin 2026-09-19).

External agreement, data.xrpl.org (UTC day 2026-04-15): 0.98832 (251 missed of 21,493). *data.xrpl.org scores UTC days from the validations its crawlers received.*

Native engine conformance

No receipt records for this day (Step 1 records begin 2026-08-29, Steps 2–5 records 2026-10-09).

Live counters are kept only for the current run; none exist for a past day.

Receipt register

No receipt records for this day (Step 1 records begin 2026-08-29).

Incidents

Not assessed: no signing records for this day.

Change log

Deployments

No deployment records before 2026-09-11.

Merged into main

Merge records begin 2026-07-04; this day's commits are listed by commit date below.

Findings fixed

No engine findings merged into main this day.

Other engineering changes

- **feat: 100k milestone push — dashboard polish + system-metrics sidecar** (`670f3c7`, xrpl-validator)
- Rolling 50-sample diverged tx buffer with L prefix (ffi_engine) - New system-metrics.py sidecar on :3779 for node-resources sparklines - Dashboard component polish: metrics-grid, node-resources, speedometer simplified (-181 net) - Wire /system/metrics rewrite in next.config.js - Ignore root-level .next/ build cache leak
- **feat(xrpl-node): wire stage3 ffi overlay path behind XRPL_FFI_STAGE3 toggle** (`d66055d`, xrpl-validator) Item L from the post-100k punch list — the Phase B scaffold. Default off; no behavioral change unless XRPL_FFI_STAGE3 is explicitly enabled.
- **fix(xrpl-node): incremental_sync hardcoded localhost + stage3 diagnostic** (`f5d7936`, xrpl-validator) Two fixes uncovered during the Stage 3 flip on the validator host: 1. incremental_sync.rs had 11 hardcoded http://localhost:5005 URLs that bypassed the XRPL_RPC_URL env var. This was a latent bug — invisible while ws_sync kept up with tip (inc-sync never ran), but fatal on any restart that left a gap. Fix: introduce a private rpc_endpoint() helper backed by a OnceLock that honors XRPL_RPC_URL and falls back to localhost …
- **feat(xrpl-node): recent_tx_samples rolling buffer for live per-tx visibility** (`214f73d`, xrpl-validator) Adds a 50-entry rolling buffer to FfiStats that captures every tx the FFI engine applies, regardless of outcome. Format: "L{ledger_seq} {tx_type}/{ter_name} {short_hash} {ms}ms mut={N}"
- **feat: stage3_enabled visibility + live TX feed in watch_engine.py** (`04a9055`, xrpl-validator)
- Add stage3_enabled: bool to FfiStats, set in start_ws_sync after reading Stage3Config. Exposed via /api/engine so dashboards and scripts can show a prominent "STAGE 3: ACTIVE" banner. - watch_engine.py: add big green

Stage 3 banner at the top (or gray "inactive" line when off), and a "Live TX Feed (last 15 applied)" section rendering recent_tx_samples with color-coded TER outcomes (green=tesSUCCESS, cyan=tec*, …

- **feat(dashboard): Stage 3 banner + Live TX Feed + hash empty state fix** (`00d29a1`, xrpl-validator) Three dashboard updates for Stage 3 visibility: 1. Big "★ Stage 3 Active ★" banner at the top of HeroBanner when ffi_verifier.stage3_enabled=true, with pulsing dot and accent glow. Gray "inactive · shadow-only Phase A" pill when off. Operators can tell at a glance which mode the validator is in.
- **feat(ffi): port shim to rippled 3.1.2 Application API** (`15840ec`, xrpl-validator) Replaces develop-era MinimalServiceRegistry with MinimalApp — a ripple::Application subclass compatible with stable rippled 3.1.2. This is the foundation for running Stage 3 against a libxrpl built from mainnet's actual running version.
- **fix(ffi): match CMakeLists.txt to rippled 3.1.2 actual dependency chain** (`d4d57ff`, xrpl-validator) ed25519 and secp256k1 are compiled from external/ subdirs, not conan packages. Import their pre-built .a files directly. Add all missing conan deps (gRPC, lz4, LibArchive, SOCI, SQLite3, RocksDB, nudb, abseil, Snappy, BZip2, ZLIB) that xrpld sources require.
- **fix(ffi): rename header() to info() for 3.1.2 ReadView interface** (`734e59b`, xrpl-validator) 3.1.2 uses info() (returning LedgerInfo const&) instead of develop's header(). LedgerInfo is a typedef for LedgerHeader so the type matches.
- **fix(ffi): use concrete OrderBookDB instead of broken NoopOrderBookDB** (`0571dc5`, xrpl-validator) 3.1.2 OrderBookDB has no virtual methods - can't subclass. Use it directly; never calling setup() keeps it empty for stateless replay.
- **fix(ffi): correct proto include path to pb-xrpl.libpb, drop stale libxrpl dir** (`9504e92`, xrpl-validator)
- **fix(ffi): add proto subdirectory include path for relative pb.h includes** (`ccbe073`, xrpl-validator)
- **fix(ffi): exclude Application.cpp (defines ApplicationImp, we use MinimalApp)** (`9bfd38b`, xrpl-validator)
- **fix(ffi): selective xrpld glob - only tx engine + dependencies** (`1ee7aac`, xrpl-validator) Exclude overlay, peerfinder, consensus, perflog, rpc directories that use incompatible boost::asio executor APIs. Include only the subsystems apply()/preflight() actually touches.
- **fix(ffi): exclude xrpld infrastructure files that pull overlay/consensus deps** (`4414f2c`, xrpl-validator) LedgerMaster, ValidatorList, NetworkOPs, PathRequests etc reference overlay Message/InfoSub symbols. Exclude them - not needed for apply.
- **fix(ffi): provide Application ctor, exclude BookListeners/Ledger, add gc-sections** (`6f7abb1`, xrpl-validator) Application::Application() was defined only in Application.cpp which we exclude. Provide it in MinimalApp.cpp. Exclude BookListeners.cpp (InfoSub deps) and Ledger.cpp (LedgerToJson deps). gc-sections strips remaining dead code that references overlay/consensus symbols.
- **fix(ffi): provide isFlagLedger definition (excluded with Ledger.cpp)** (`6633d5c`, xrpl-validator) Simple seq % 256 check needed by Change.cpp pseudo-tx handler.
- **fix(ffi): add ed25519/secp256k1 link paths in build.rs** (`8386d98`, xrpl-validator) These are built from rippled's external/ dirs, not conan packages. Need explicit link search paths + lib names for Rust linker.
- **fix(ffi): exclude conan OpenSSL from Rust link (version conflict with system 3.x)** (`1697fad`, xrpl-validator) Conan provides OpenSSL 1.1.x static libs. The Rust openssl crate links system OpenSSL 3.x dynamically. Mixing them causes missing 3.x-only symbols. Let Rust handle its own OpenSSL linking.
- **fix(ffi): skip entire OpenSSL conan package in build.rs** (`fc1947d`, xrpl-validator) Excluding just lib names wasn't enough - the conan OpenSSL search paths cause the linker to find 1.1.x .a files before system 3.x .so. Skip the OpenSSL cmake data file entirely.

- **fix(ffi): use FFI drops_destroyed for fee tracking instead of byte scanner** (`2731504`, xrpl-validator) extract_fee_drops was a byte-pattern scanner that matched 0x68 0x4_ in raw tx blobs. False positives inside signatures produced absurd fee values (10^19 drops). Use the FFI apply result's drops_destroyed which comes from libxrpl's actual transaction processing.
- **fix(dashboard): remove fake validator agreement percentages** (`3bc9c5a`, xrpl-validator) The consensus grid had hardcoded fake lower percentages for V14, V28, V32, V34 etc. that made them appear non-green. Replace with real active/inactive state based on actual agreement count from backend.
- **fix(dashboard): update validation target to 50K, rename to Validation Window** (`6d25337`, xrpl-validator)
- **fix(sync): abort write batch when any SLE fetch fails** (`e9e3fbc`, xrpl-validator) Previously, process_ledger wrote all successfully-fetched SLEs to state.rocks even when some fetches failed. The missing keys silently corrupted the state, causing hash mismatches ledgers later when transactions touched the missing entries.
- **fix(dashboard): use shadow_hash_attempted for ledger count and progress bar** (`911bd91`, xrpl-validator) The validation window tracks verified shadow hash comparisons, not raw FFI engine ledger count. Renamed label to "Verified".
- **feat(ffi): add persistent shadow hash mismatch logger** (`5c90319`, xrpl-validator) Writes to logs/shadow_mismatches.jsonl (or XRPL_FFI_SHADOW_LOG env). Each entry records timestamp, ledger seq, our hash, network hash, and overlay key count. Same JSONL pattern as divergences.jsonl.
- **fix(ffi): get ledger seq from stats for shadow mismatch logger** (`9540234`, xrpl-validator)

Tests and campaigns

- **test(xrpl-ffi): TicketCreate harness tests for the post-100k divergence hunt** (`a62144b`, xrpl-validator)
- **test(xrpl-node): apply_ledger_in_order reproducer for TicketCreate cascade** (`448ba95`, xrpl-validator)

Docs and operations

- **chore: update libxrpl version references from 3.2.0-b0 to 3.1.2** (`360cae9`, xrpl-validator)
- **docs: why libxrpl was downgraded from develop (3.2.0-b0) to stable (3.1.2)** (`879f389`, xrpl-validator)
- **diag(ffi): dump FFI vs AffectedNodes overlay diff on shadow mismatch** (`5be9359`, xrpl-validator)

Independence program

No independence record for UTC day 2026-04-15; records begin 2026-10-20.

Network

No ledger records for this day: the reference node holds ledgers from 2026-10-04 22:52 UTC (#107,436,647).

Amendment watch

No amendment records for this day.

Open items

None.

Corrections to earlier reports

None.

Appendix — commits

Committed	Merged into main	Repo	Commit	Subject
04-15 11:21	—	xrpl-validator	`670f3c7`	feat: 100k milestone push — dashboard polish + system-metrics sidecar
04-15 11:41	—	xrpl-validator	`d66055d`	feat(xrpl-node): wire stage3 ffi overlay path behind XRPL_FFI_STAGE3 toggle
04-15 12:23	—	xrpl-validator	`a62144b`	test(xrpl-ffi): TicketCreate harness tests for the post-100k divergence hunt
04-15 13:14	—	xrpl-validator	`448ba95`	test(xrpl-node): apply_ledger_in_order reproducer for TicketCreate cascade
04-15 23:53	—	xrpl-validator	`f5d7936`	fix(xrpl-node): incremental_sync hardcoded localhost + stage3 diagnostic
04-16 00:06	—	xrpl-validator	`214f73d`	feat(xrpl-node): recent_tx_samples rolling buffer for live per-tx visibility
04-16 00:12	—	xrpl-validator	`04a9055`	feat: stage3_enabled visibility + live TX feed in watch_engine.py
04-16 00:28	—	xrpl-validator	`00d29a1`	feat(dashboard): Stage 3 banner + Live TX Feed + hash empty state fix
04-16 00:54	—	xrpl-validator	`15840ec`	feat(ffi): port shim to rippled 3.1.2 Application API
04-16 01:17	—	xrpl-validator	`d4d57ff`	fix(ffi): match CMakeLists.txt to rippled 3.1.2 actual dependency chain
04-16 01:19	—	xrpl-validator	`734e59b`	fix(ffi): rename header() to info() for 3.1.2 ReadView interface
04-16 01:21	—	xrpl-validator	`0571dc5`	fix(ffi): use concrete OrderBookDB instead of broken NoopOrderBookDB
04-16 01:23	—	xrpl-validator	`9504e92`	fix(ffi): correct proto include path to pb-xrpl.libpb, drop stale libxrpl dir
04-16 01:24	—	xrpl-validator	`ccbe073`	fix(ffi): add proto subdirectory include path for relative pb.h includes
04-16 01:25	—	xrpl-validator	`9bfd38b`	fix(ffi): exclude Application.cpp (defines ApplicationImp, we use MinimalApp)
04-16 01:29	—	xrpl-validator	`1ee7aac`	fix(ffi): selective xrpld glob - only tx engine + dependencies
04-16 01:31	—	xrpl-validator	`4414f2c`	fix(ffi): exclude xrpld infrastructure files that pull overlay/consensus deps
04-16 01:33	—	xrpl-validator	`6f7abb1`	fix(ffi): provide Application ctor, exclude BookListeners/Ledger, add gc-sections
04-16 01:36	—	xrpl-validator	`6633d5c`	fix(ffi): provide isFlagLedger definition (excluded with Ledger.cpp)

Committed	Merged into main	Repo	Commit	Subject
04-16 01:38	—	xrpl-validator	`8386d98`	fix(ffi): add ed25519/secp256k1 link paths in build.rs
04-16 01:39	—	xrpl-validator	`1697fad`	fix(ffi): exclude conan OpenSSL from Rust link (version conflict with system 3.x)
04-16 01:40	—	xrpl-validator	`fc1947d`	fix(ffi): skip entire OpenSSL conan package in build.rs
04-16 01:53	—	xrpl-validator	`2731504`	fix(ffi): use FFI drops_destroyed for fee tracking instead of byte scanner
04-16 01:59	—	xrpl-validator	`360cae9`	chore: update libxrpl version references from 3.2.0-b0 to 3.1.2
04-16 02:01	—	xrpl-validator	`879f389`	docs: why libxrpl was downgraded from develop (3.2.0-b0) to stable (3.1.2)
04-16 02:04	—	xrpl-validator	`3bc9c5a`	fix(dashboard): remove fake validator agreement percentages
04-16 02:07	—	xrpl-validator	`6d25337`	fix(dashboard): update validation target to 50K, rename to Validation Window
04-16 02:13	—	xrpl-validator	`e9e3fbc`	fix(sync): abort write batch when any SLE fetch fails
04-16 02:19	—	xrpl-validator	`911bd91`	fix(dashboard): use shadow_hash_attempted for ledger count and progress bar
04-16 02:22	—	xrpl-validator	`5c90319`	feat(ffi): add persistent shadow hash mismatch logger
04-16 02:24	—	xrpl-validator	`9540234`	fix(ffi): get ledger seq from stats for shadow mismatch logger
04-16 02:49	—	xrpl-validator	`5be9359`	diag(ffi): dump FFI vs AffectedNodes overlay diff on shadow mismatch

Times in UTC. A commit appears on the day it was committed to main or merged into it.